Experiment - 2

Aim -: Convert the ER-diagram into a relational schema write complete mysql create table statement with primary key, foreign key, Not null, unique, delete, cascade and set null constraints. Insert sample data & demonstrate referential integrity.

Source code -:

```
Create database E-commerce;
Use E-commerce;
Create table customer (
        customer_ID int primary key;
        name varchar(100) not null;
        email varchar (100) unique;
        age int);
Create table seller (
        ID int primary key;
        phone int;
        Email varchar (100)
    );
Create table category (
        ID int primary key;
        name varchar (100) not null;
    );
```

```
Create table Product (
    Product_id int primary key,
    name varchar (100) not null,
    Quantity int,
    Price not null,
    Category_id int ,
    Foreign key (Category_id ) References category (category_id)
    On delete Cascade
);

Create Table Payment(
    Pa_id int primary key,
    amount decimal (10,2) not null,
    Pa_date date,
    method varchar (50),
);

Create table Delivery (
    address_ int primary key,
    City varchar (50)
    date varchar (50)
    Pincode varchar (10)
);

Create table order
    order_id int Primary key;
    order_date date;
    amount decimal (10,2)
    seller_id int;
    address_id int;
```

```
    Foreign key (customer_id) Reference customer (customed_id)
    on delete cascade;
    Foreign key (seller_id) Reference seller (seller_id)
    Foreign key (Payment_id) Reference Payment (payment_id)
    Foreign key (address_id) Reference Delivery_address (address_id)
);

Insert into customer values (1, 'Rahul', 'rahul@gmail.com, 21),
(2, 'Aman', 'aman@gmail.com', 22), (3, 'Om', 'om@gmail.com, 23);

Insert into seller values (1, 'Tech', 'Delhi') (2, 'Fashion hub', 'Uko')
(3, 'Gadget Zone', 'Banglore');

Insert into category values (1, 'Electronic', 'Electonic Product')
(2, 'Clothing', 'Clothes & Fashion') (3, 'Home', 'Home appliances');

Insert into Product values (101, 'laptop', 10, 50, 0.00) (102, 'T-shirt',
27,799, 2.01) (103, 'DBMS Books', 20,000);

Insert into payment values (201, 55000.0, '2026-09-01', 'UPI')
(202, 799.00, '2026-09-02', 'UPI') (205, '1200.00', '2026-09-05', 'Net Banking
```

Insert into delivery address values (301, 'Gurga', 'Haryana', '2201')
(302, 'mumbai', 'Maharashtra', 40001) (303, 'Jaipur', 'Rajasthan', 30400)

Insert into order Value (401, '2026-09-01', 55,000.00, 1, 1, 201, 301).
(402, '2026-09-02', 799.00, 2, 2, 302, 303)
(403, '2026-09-03', 3500.00, 3, 3, 203, 303)
(404, '2026-09-04', 650.00, 4, 4, 204, 304)

Outcome -:

1. Customer

Customer_id	Name	email	age
1	Rahul	rahul@gmail.com	21
2	Aman	Aman@gmail.com	23
3	Om	om@gmail.com	23

2. Seller

seller_id	name	seller : address
1	Tech	Delhi
2	Fashion	Mumbai
3	Home	LKO

3. Category

Id	Ca_name	description
1	electronic	electronic Product
2	clothing	clothes & fashion
3	Home	Home appliances.

4. Product

Pr_id	name	quantity	Price	category_id
101	Laptop	10	5500.00	1
102	T-shirt	15	799.00	2
103	DBMS Books	20	650.00	5

5. Payment

Pay_id	amount	date	method
201	5500.00	2026-09-01	UPI
202	799.00	2026-09-02	UPI
203	650.00	2026-09-03	net Banking

6 Delivery address

address_id	City	State	Pincode.
301	Gurgaon	Haryana	122061
302	Mumbai	Maharashtra	400001
303	Pune	UP	226016

7. Orders

Order_id	order_date	amount	cus_id	seller_id
401	2026-09-01	5500.00	1	1
402	2026-09-02	749.00	2	2
403	2026-09-02	650.00	4	4

Pay_id	add_id
201	301
202	302
203	303

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